



FASR Science Case # WG2-1

# Mapping Magnetic Fields in Post-Flare Loops and Arcades

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## 1 Science Goal(s)

*Briefly summarize the key science goal(s) for this science case. A few sentences will be sufficient.*

The primary goal of this science case is to measure and map magnetic-field strength in post-flare loops and arcades after magnetic reconnection during solar flares. The main discrimination test is whether the field inferred from spatially resolved microwave spectra and circular polarization follows the formation, rise, and cooling of newly reconnected loops. The fits must separate changes in field strength from changes in the energetic-electron distribution, so that a brightening can be interpreted as reconnection and particle transport rather than as a change in viewing geometry alone.

## 2 Scientific Rationale

### 2.1 Scientific Importance

*Provide a brief discussion on the scientific importance of this science case.*

Post-flare loops and arcades are direct products of magnetic reconnection. Their magnetic-field strength controls the gyrosynchrotron spectrum, particle trapping, energy transport, and the evolution of the newly reconnected flare system. Quantitative coronal magnetic-field measurements in these structures test how magnetic energy is converted into accelerated electrons, heating, and radiation after reconnection. They also provide field constraints for interpreting post-flare arcades in eruption and space-weather studies; this case does not by itself claim predictive flare or CME capability.

### 2.2 Uniqueness to FASR Capabilities

*Is this science case uniquely addressed by FASR? Why can't other facilities address this science and achieve the same goal?*

FASR offers a unique capability for post-flare arcade magnetometry because it can provide broadband imaging spectroscopy and circular polarization over the 2-20 GHz range at high cadence. Gyrosynchrotron spectra from flare-accelerated electrons encode magnetic-field strength, viewing geometry, and non-thermal electron properties. Stokes I provides the spatially resolved spectrum, while circular polarization provides an additional diagnostic of the coronal magnetic field. FASR

therefore uniquely probes magnetic-field strength and topology inside flare loops and arcades during their rapid evolution.

## 2.3 Synergies

*Describe potential synergies/complementarities between this FASR science case and those from current/future/planned facilities at all wavelengths (e.g., DKIST, MUSE, FIERCE, COSMO, ngGONG, etc.).*

FASR post-flare arcade measurements should be interpreted within a broader multi-wavelength context. AIA, SUVI, MUSE, and related EUV/UV imagers and spectrographs can trace loop morphology, plasma flows, cooling, and temperature structure. DKIST, COSMO, ngGONG, and related magnetic-field facilities can provide photospheric, chromospheric, and off-limb coronal magnetic constraints. In particular, infrared coronal diagnostics can provide line-of-sight magnetic-field measurements for flare arcades near the limb, where scattering geometry and line formation are favorable. Hard X-ray observations can identify higher-energy electron precipitation and looptop acceleration sites. In contrast, microwave gyrosynchrotron emission allows FASR to measure magnetic-field strength and direction in flare loops and arcades regardless of flare location on the solar disk. Together, these observations place the FASR microwave magnetic-field measurements into a comprehensive flare energy-release and coronal magnetic-field context.

## 2.4 Measurements Required by FASR

*Provide a description of the necessary measurements to be carried out by the FASR to adequately address this science case. Please coordinate these measurements with the Science Requirements table in Section III.*

The inference uses the turnover and optically thin slope of the gyrosynchrotron spectrum, together with (V/I), to recover magnetic-field strength and electron parameters separately in loop-top, loop-leg, and arcade regions. The baseline product is a spatially registered Stokes I/V spectrum covering 2–20 GHz every 5 s; a 1 s product is a goal for the impulsive phase. Stokes Q/U are optional and do not control the baseline.

## III. Science Requirements Tables

## (A) Observational Target Description

*Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.*

| (A) OBSERVATIONAL TARGET                         |                                                                                                                                          |                                                                                     |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Type of observation<br>(what defines a ‘target’) | Provide a brief description of the target:<br>Gyrosynchrotron emission from flare-accelerated electrons in post-flare loops and arcades. |                                                                                     |
| Number of targets                                | One to a few post-flare arcade systems per event.                                                                                        |                                                                                     |
| Size of a single target (arcsec x arcsec)        | < 10''-100s''                                                                                                                            |                                                                                     |
| Distribution of all targets (arcmin x arcmin)    | < 30 arcmin across the solar disk                                                                                                        |                                                                                     |
| Peak brightness (sfu/beam or Kelvin)             | $10^6$ – $10^{10}$ K, depending on flare magnitude and frequency                                                                         |                                                                                     |
| RMS brightness (sfu/beam or Kelvin)              | $10^5$ K per spectral sample; this gives a 10-sigma detection of a $10^6$ K extended loop source.                                        |                                                                                     |
| Expected circularly polarized flux density       | Stokes V<br>(sfu/beam)                                                                                                                   |                                                                                     |
|                                                  | V/I                                                                                                                                      | 10%–50% in the optically thin source; the sign and spectral variation are required. |
| Expected linearly polarized flux density         | Stokes Q or U<br>(sfu/beam)                                                                                                              | N/A                                                                                 |
|                                                  | Q/I or U/I                                                                                                                               | N/A                                                                                 |

## (B) Spectro-Temporal Requirements Discussion

*Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.*

| (B) SPECTRAL-TEMPORAL REQUIREMENTS |                                                                                                                                                            |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Central Frequency (GHz)            | 2 to 20 baseline; extension above 20 GHz is a goal for events whose turnover remains high.                                                                 |
| Instantaneous Bandwidth (GHz/pol)  | The observing sequence must sample both sides of the spatially varying spectral turnover within the stated cadence; backend allocation is a design choice. |
| Spectral resolution [MHz]          | $\Delta\nu/\nu \leq 2\%$ baseline (about 100 MHz at 5 GHz); finer sampling is a goal if required by injection/recovery tests.                              |
| Temporal resolution (in seconds)   | 5 baseline; 1 goal during the impulsive phase.                                                                                                             |

## (C) Polarization Product Discussion

*Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.*

| (C) POLARIZATION DATA PRODUCTS REQUIRED |          |
|-----------------------------------------|----------|
| (y)                                     | Stokes I |
| (n)                                     | Stokes Q |
| (n)                                     | Stokes U |
| (y)                                     | Stokes V |

## (D) Imaging Requirements Discussion

*Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.*

The imaging requirements are set by the need to resolve post-flare loop and arcade magnetic fields, rather than by the need to image every structure in an eruption. The central science questions are: how does magnetic-field strength vary between loop-top, loop-leg, and arcade sources; how does it evolve as newly reconnected loops form and cool; and how do the spatially resolved microwave spectrum and circular polarization constrain the energetic-electron population and coronal magnetic field within those loops? More broadly, FASR must determine the spatial distribution and temporal evolution of coronal magnetic-field strength and topology in the flare core and surrounding arcade, identify compact sites of reconnection and particle acceleration, and establish how these regions are magnetically connected to the larger erupting structure. The instrumental requirements discussed below follow directly from these science goals and therefore provide the basis for the relevant system-design considerations.

Angular resolution is set by the need to distinguish compact footpoint or loop-leg sources from broader looptop and arcade emission. A simple back-of-the-envelope estimate can be obtained from magnetic flux conservation along a loop,  $B_{\text{fp}}A_{\text{fp}} \approx B_{\text{lt}}A_{\text{lt}}$ . If the source cross-section has a characteristic linear size  $w$ , this gives  $w_{\text{lt}} \approx w_{\text{fp}}(B_{\text{fp}}/B_{\text{lt}})^{1/2}$ . For arcsecond-scale footpoints with  $w_{\text{fp}} \sim 1''\text{--}3''$ , rooted in fields  $B_{\text{fp}} \sim 500\text{--}1500$  G, and for looptop fields  $B_{\text{lt}} \sim 50\text{--}150$  G, the expected looptop size is typically a few to about 10 arcsec. This implies that arcsecond-class resolution is needed at the high-frequency end to resolve compact strong-field sources, while several-arcsecond resolution is needed at lower frequencies to separate and characterize looptop and arcade emission.

The mapped image size is driven by the need to image not only the bright flare kernel but the full active-region magnetic system, including surrounding arcades and connections to erupting structures. A mapped field as large as about 10 arcmin, roughly one-third of the solar-disk diameter, provides a conservative system-level envelope for preserving that context.

The required pixel scale is a derived imaging choice rather than a primary science requirement. To recover morphology and polarization structure reliably, the synthesized beam should be sampled by roughly 3-5 pixels across the FWHM. Thus a  $10''$  beam at low frequency implies pixels of about  $2''\text{--}3''$ , while a  $1''$  beam at high frequency implies pixels of about  $0.2''\text{--}0.3''$ . This gives a more direct science-based interpretation of the pixel-size requirement than treating pixel scale as an independent quantity.

Spectral imaging requirements are set by the physics of the spatially varying gyrosynchrotron continuum. The turnover and low-frequency curvature encode magnetic field strength, characteristic electron energy, and viewing angle, while the optically thin slope and (V/I) provide complementary constraints (Dulk 1985; Marsh et al. 1980). The science acceptance test is recovery of these parameters for injected loop-top, loop-leg, and arcade spectra whose turnovers vary across the source and evolve during the observation. The current 2 percent fractional sampling is a baseline starting point; a design-phase recovery study must determine whether it recovers the field and electron parameters within the adopted errors. The sampling is therefore specified fractionally and is not tied to a fixed channel width or data-volume compromise.

Sensitivity and dynamic range are set by recovery of weak, spatially extended loop and arcade emission in the presence of much brighter compact flare sources. The baseline acceptance test injects representative loop-top, loop-leg, and arcade spectra into realistic flare fields and requires recovery of a  $10^6$  K extended source at signal-to-noise of at least 10, with field and electron-parameter errors reported separately from source self-noise, sidelobes, and calibration bias. The  $10^5$  K value is therefore a science-motivated starting point per fractional spectral sample, not a fixed 100 MHz backend requirement; the final threshold follows from the parameter-recovery test (Gary et al. 2018; Chen et al. 2020; Bastian et al. 2025).

Polarization accuracy is required because (V/I) is part of the gyrosynchrotron fit. Published flare sources commonly show 10–50 percent circular polarization, so a 5–10 percent relative accuracy on (V/I) is an initial baseline for recovering its sign and spectral trend; injection/recovery tests must establish whether the weakest accepted sources require a tighter value (Dulk 1985; Marsh et al. 1980). Q/U are not needed for this case because the science inference uses circular polarization and total intensity.

In summary, the science case drives FASR toward a broadband imaging spectropolarimeter with frequency-dependent spatial resolution, sensitivity to both compact and extended structures, stable polarization calibration, and enough spectral and temporal coverage to follow the rapid evolution of flare-related coronal magnetic fields.

| <b>(D) IMAGING REQUIREMENTS</b>                                 |                                                                                                      |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Required angular resolution (arcsec)<br>(single value or range) | 10'' at 2 GHz                                                                                        |
| Largest angular scale required (arcsec)                         | At least (600'') to retain the arcade and active-region background used in the spectral fit.         |
| Mapped image size (arcmin x arcmin)                             | Up to 10' × 10'                                                                                      |
| Required pixel resolution (arcsec)                              | 6'' at 1 GHz and 0.3'' at 20 GHz                                                                     |
| Number of output/image channels                                 | Set by $\Delta\nu/\nu \leq 2\%$ over the selected band; approximately 150–200 samples over 2–20 GHz. |
| Output bandwidth (minimum and maximum frequency - GHz)          | 2–20 baseline; extension above 20 GHz is a goal.                                                     |
| Channel width (MHz)                                             | Fractional width $\leq 2\%$ , rather than a fixed backend channel count.                             |

| (D) IMAGING REQUIREMENTS (continued)                                                                |                   |
|-----------------------------------------------------------------------------------------------------|-------------------|
| Required rms (sfu/beam or Kelvin) [per channel] (if polarization products required define for each) | 10 <sup>5</sup> K |
| Dynamic range within image (if polarization products required define for each)                      | 1000:1            |
| Polarization accuracy (%)                                                                           | 5-10%             |
| Zero spacing/total power required?                                                                  | y                 |
| Required maximum latency (in seconds, or N/A)                                                       | N/A               |
| Required flux density scale calibration accuracy                                                    | 1-3%              |
|                                                                                                     | 5%                |
|                                                                                                     | x 10%             |
|                                                                                                     | 20-50%            |

## 2.5 Other Performance or Functional Requirements

*If there are any additional performance or functional requirements not captured above, describe them here. For example, beamforming array mode, phased array, etc.*

## 3 Appendix

*Please provide any other relevant material necessary to understand and substantiate this Science Case.*

## Reference

- Tomczyk, S., E. Landi, J. T. Burkepile, et al. 2016. “Scientific Objectives and Capabilities of the Coronal Solar Magnetism Observatory.” *Journal of Geophysical Research (Space Physics)* 121 (August): 7470–87. <https://doi.org/10.1002/2016JA022871>.
- Dulk, G. A. 1985. “Radio Emission from the Sun and Stars.” *Annual Review of Astronomy and Astrophysics* 23: 169–224. <https://doi.org/10.1146/annurev.aa.23.090185.001125>.
- Gary, D. E., B. Chen, B. R. Dennis, G. D. Fleishman, G. J. Hurford, S. Krucker, J. M. McTiernan, et al. 2018. “Microwave and Hard X-Ray Observations of the 2017 September 10 Solar Limb Flare.” *The Astrophysical Journal* 863: 83. <https://doi.org/10.3847/1538-4357/aad0ef>.
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- Chen, B., C. Shen, D. E. Gary, K. K. Reeves, G. D. Fleishman, S. Yu, F. Guo, et al. 2020. “Measurement of Magnetic Field and Relativistic Electrons Along a Solar Flare Current Sheet.” *Nature Astronomy* 4: 1140–1147. <https://doi.org/10.1038/s41550-020-1147-7>.
- Kou, Y., X. Cheng, Y. Wang, S. Yu, B. Chen, E. P. Kontar, et al. 2022. “Microwave Imaging of Quasi-Periodic Pulsations at Flare Current Sheet.” *Nature Communications* 13: 7680. <https://doi.org/10.1038/s41467-022-35377-0>.

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